ARNAV ATUL WAGHDHARE

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EDUCATION

- **KJ Somaiya College of Engineering** *August 2023 - Present*
  B.Tech in Computer Engineering Mumbai, India
  ◦ CGPA: 8.84

EXPERIENCE

- **Full Stack Developer** *May 2026 - August 2026*
  Datafuel Mumbai, India
  ◦ Built high-throughput Python data pipelines using **multithreading/multiprocessing** to scrape **1,000+ darkstores** in 4 hours, generating **15M+ records** in ClickHouse while bypassing anti-bot systems.
  ◦ Engineered end-to-end **web scraping & ETL pipelines** by reverse engineering private APIs, and developing **ReactJS** analytics dashboards for enterprise clients for asynchronous task management.

- **Data Scientist Intern** *July 2025 - August 2025*
  Evoastra Ventures Remote
  ◦ **Led my team** to build and deploy a web scraper for dynamic sites, handling large-scale data collection.
  ◦ Built a comprehensive database of car information and pricing, using **Selenium** and **BeautifulSoup**.
  ◦ Developed a CNN+LSTM image captioning model in **TensorFlow** with a BLEU score of **0.56**.

PROJECTS

- **Astra: Real-Time Streaming Voice Assistant** *July 2026 - September 2026*
  Tools: Go, Python, ONNX Runtime, PyTorch, faster-whisper, Piper TTS, OpenWakeWord, gRPC, WebSocket, Docker [GitHub]
  ◦ Engineered a real-time voice pipeline in **Go** (20ms PCM over **WebSocket/protobuf**, **gRPC** fan-out to **faster-whisper** ASR and **Piper TTS**), cutting E2E Time-To-First-Audio from **5924ms to 3765ms**.
  ◦ Trained custom **VAD** (**AUC 0.982**, **11.6 false alarms/hr**) and **OpenWakeWord** wake-word (**0.96 recall**) models in **PyTorch**, deployed to Go via **ONNX**. Implemented server-side barge-in and **Docker Compose** orchestration.

- **Gesture Scrolling: Hand Gesture Recognition for Scrolling on Android** *June 2025 - July 2025*
  Tools: PyTorch, OpenCV, Mediapipe, Optuna, ONNX, Kotlin, Android Studio [GitHub, GitHub]
  ◦ Designed and trained a custom **Temporal Convolutional Network (TCN)** model, tuning hyperparameters with **Optuna** to raise accuracy **(91.7% to 95.8%)** and utilizing **ONNX** for **8.23x** inference speedup.
  ◦ Shipped a **Kotlin/Jetpack Compose** Android app for hands-free social-media navigation from a live camera feed.

- **AI Powered Malware Detection and Classification System** *April 2025*
  Tools: PyTorch, Kaggle, SciKit-Learn, NodeJS, ExpressJS [GitHub]
  ◦ Engineered a **hybrid ML model** during a 24-hour hackathon by training a **CNN and XGBoost** pipeline.
  ◦ Built a real-time Python inference pipeline classifying malware from bytecode images, integrated with a **NodeJS/ExpressJS** backend.

SKILLS

- **Programming Languages:** Python, C++, Java, Go, JavaScript, SQL, Kotlin, Shell, HTML, CSS
- **Web & Databases:** ReactJS, NodeJS, FastAPI, REST APIs, gRPC, WebSocket, Protocol Buffers, Celery, TailwindCSS, AJAX, PostgreSQL, AWS Aurora, Git, Docker
- **Data Science & Machine Learning:** PyTorch, Lightning, TensorFlow, ONNX Runtime, faster-whisper, Piper TTS, OpenWakeWord, LangChain, NumPy, Pandas, Scikit-learn, Google-ADK, Ollama, OpenRouter
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

- **DeepLearning.ai:** Supervised Machine Learning: Regression and Classification *November 2024*
- **DeepLearning.ai:** Advanced Learning Algorithms *December 2024*
- **IUCEE:** Mastering Multi-Agent AI: Building Intelligent Collaborative System *December 2024*
- **GAJSHIELD 2025:** Ranked **Top 5** in the GAJSHIELD Hackathon *April 2025*